15. If X and Y are two variates, there can be atmost
    (a) One regression line
    (b) two regression lines
    (c) three regression lines
    (d) more regression lines

16. The lines of regression of X on Y estimates
    (a) X for a given value of Y
    (b) Y for a given value of X
    (c) X from Y and Y from X
    (d) none of these

17. Scatter diagram of the variate values (X,Y) give the idea about
    (a) functional relationship
    (b) regression model
    (c) distribution of errors
    (d) no relation

18. If regression co-efficient of Y on X is 2, then the regression co-efficient of X on Y is
    (a) $\leq \frac{1}{2}$ (b) 2
    (c) $> \frac{1}{2}$ (d) 1

19. If two variables moves in decreasing direction then the correlation is
    (a) positive (b) negative
    (c) perfect negative (d) no correlation

20. The person suggested a mathematical method for measuring the magnitude of linear relationship between two variables say X and Y is
    (a) Karl Pearson
    (b) Spearman
    (c) Croxton and Cowden
    (d) Ya Lun Chou

21. The lines of regression intersect at the point
    (a) (X,Y) (b) $(\overline{X}, \overline{Y})$
    (c) $(0,0)$ (d) (σ_x, σ_y)

22. The term regression was introduced by
    (a) R.A. Fisher
    (b) Sir Francis Galton
    (c) Karl Pearson
    (d) Croxton and Cowden

23. If $r=-1$, then correlation between the variables
    (a) perfect positive
    (b) perfect negative
    (c) negative
    (d) no correlation

24. The coefficient of correlation describes
    (a) the magnitude and direction
    (b) only magnitude
    (c) only direction
    (d) no magnitude and no direction

25. If $\text{Cov}(x,y)=-16.5$, $\sigma_x^2=2.89$, $\sigma_y^2=100$. Find correlation coefficient.
    (a) –0.12 (b) 0.001
    (c) –1 (d –0.97

Miscellaneous Problems

1. Find the coefficient of correlation for the following data:

X	35	40	60	79	83	95
Y	17	28	30	32	38	49

2. Calculate the coefficient of correlation from the following data:

   $\Sigma X=50$, $\Sigma Y=-30$, $\Sigma X^2=290$, $\Sigma Y^2=300$, $\Sigma XY=-115$, $N=10$

3. Calculate the correlation coefficient from the data given below:

X	1	2	3	4	5	6	7	8	9
Y	9	8	10	12	11	13	14	16	15

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